SHORT · COMPUTE CHOICE

Choose Databricks compute by workload

Choose Databricks compute by the workload first, not by cluster size. Interactive notebooks use serverless notebook compute, or classic all-purpose compute when you need a custom configuration. Scheduled production uses serverless jobs or classic jobs compute. Structured Query Language (SQL) analytics, business intelligence (BI) tools, and dashboards use a SQL warehouse, ideally serverless for a fast start. If classic jobs must start fast, back them with an instance pool.

Key points

- Interactive notebooks: serverless notebook compute, or all-purpose compute for a custom size or runtime.
- Scheduled production: serverless jobs, or classic jobs compute, which has a lower Databricks Unit (DBU) rate than all-purpose.
- SQL, BI, and dashboards: a SQL warehouse, ideally serverless.
- Fast classic start: attach an instance pool. Pools do not charge DBUs while idle, but the cloud provider still bills the instances.
- A DBU is a Databricks Unit of processing capability, billed per second of use. Classic compute also bills virtual machine (VM) cost from your cloud provider. Serverless DBU price covers the compute that runs in the Databricks account.

Tiny example

When you read a certification scenario, underline the workload words first, then pick the compute.

```
interactive  ->  serverless notebooks  (or all-purpose if custom)
scheduled    ->  serverless jobs or classic jobs compute
SQL / BI     ->  SQL warehouse, ideally serverless
fast start   ->  back classic jobs with a pool
```

Misconception check

All-purpose compute is not the default for scheduled production. Jobs compute has a lower DBU rate, so daily extract, transform, load (ETL) belongs on jobs compute or serverless jobs, not on an interactive cluster.

Memory trick

When you read a scenario, look for the workload words first. Interactive, scheduled, SQL, startup time, and cost.

Remember

Pick compute by the workload. Notebooks explore, jobs produce, warehouses answer, and pools start classic jobs faster.

Validated against Databricks Free Edition limits: serverless notebook compute and one 2X-Small SQL warehouse only. Classic all-purpose compute, jobs compute, and instance pools are exam concepts here. They are not available on Free Edition.